

2014 年数学(三)真题解析

一、选择题

(1) 【答案】 (A)

【解】 由 $\lim_{n \rightarrow \infty} a_n = a$ 得 $\lim_{n \rightarrow \infty} |a_n| = |a|$,

取 $\varepsilon_0 = \frac{|a|}{2}$, 总存在 $N > 0$, 当 $n > N$ 时,

$||a_n| - |a|| < \frac{|a|}{2}$, 从而 $\frac{|a|}{2} < |a_n| < \frac{3|a|}{2}$, 应选(A).

(2) 【答案】 (C)

【解】 对 $y = x + \sin \frac{1}{x}$,

由 $\lim_{x \rightarrow \infty} \frac{y}{x} = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \sin \frac{1}{x}\right) = 1$,

$\lim_{x \rightarrow \infty} (y - x) = \lim_{x \rightarrow \infty} \sin \frac{1}{x} = 0$

得曲线 $y = x + \sin \frac{1}{x}$ 有斜渐近线 $y = x$, 应选(C).

(3) 【答案】 (D)

【解】 方法一 令 $f(x) = p(x) - \tan x$,

因为 $p(x) - \tan x$ 是比 x^3 高阶的无穷小, 所以 $f(0) = f'(0) = f''(0)$,

而 $f'(x) = b + 2cx + 3dx^2 - \sec^2 x$, $f''(x) = 2c + 6dx - 2\sec^2 x \tan x$,

则 $a = 0, b = 1, c = 0$, 应选(D).

方法二 因为 $\tan x = x + \frac{1}{3}x^3 + o(x^3)$,

所以 $p(x) - \tan x = a + (b-1)x + cx^2 + \left(d - \frac{1}{3}\right)x^3 + o(x^3)$,

于是 $a = 0, b = 1, c = 0, d = \frac{1}{3}$, 应选(D).

方法三 由 $\lim_{x \rightarrow 0} [p(x) - \tan x] = 0$ 得 $a = 0$;

$p(x) - \tan x = bx + cx^2 + dx^3 - \tan x$,

由 $\lim_{x \rightarrow 0} \frac{p(x) - \tan x}{x} = 0$ 得 $b - 1 = 0$, 即 $b = 1$;

由 $\lim_{x \rightarrow 0} \frac{p(x) - \tan x}{x^2} = \lim_{x \rightarrow 0} \frac{(x - \tan x) + cx^2 + dx^3}{x^2} = 0$ 得 $c = 0$;

由 $\lim_{x \rightarrow 0} \frac{p(x) - \tan x}{x^3} = \lim_{x \rightarrow 0} \frac{(x - \tan x) + dx^3}{x^3} = \lim_{x \rightarrow 0} \frac{1 - \sec^2 x}{3x^2} + d = d - \frac{1}{3} = 0$ 得 $d = \frac{1}{3}$,

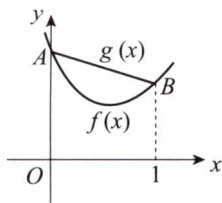
应选(D).

(4) 【答案】 (D).

【解】 方法一 令 $\varphi(x) = f(x) - g(x) = f(x) - f(0)(1-x) - f(1)x$ 且 $\varphi''(x) = f''(x)$,

当 $f''(x) \geq 0$ 时, $\varphi''(x) = f''(x) \geq 0$, 曲线 $y = \varphi(x)$ 为凹函数, 因为 $\varphi(0) = 0, \varphi(1) = 0$, 所以当 $x \in [0, 1]$ 时, $\varphi(x) \leq 0$, 即 $f(x) \leq g(x)$, 应选(D).

方法二 当 $f''(x) \geq 0$ 时, $y = f(x)$ 为凹函数, 因为 $y = g(x)$ 在 $[0, 1]$ 上为连接 $A(0, f(0))$ 与 $B(1, f(1))$ 的线段, 所以 $f(x) \leq g(x)$, 应选(D).



—(4) 题图

方法点评: 本题考查函数大小比较.

利用凹凸性证明不等式是不等式证明的重要方法, 设函数 $f(x)$ 在 $[a, b]$ 上二阶可导, 且 $f''(x) \geq 0 (\leq 0)$, 若 $f(a) = f(b) = 0$, 则当 $x \in [a, b]$ 时, $f(x) \leq 0 (\geq 0)$.

(5) **【答案】** (B).

$$\begin{aligned} \text{【解】} \quad & \begin{vmatrix} 0 & a & b & 0 \\ a & 0 & 0 & b \\ 0 & c & d & 0 \\ c & 0 & 0 & d \end{vmatrix} = -a \begin{vmatrix} a & 0 & b \\ 0 & d & 0 \\ c & 0 & d \end{vmatrix} + b \begin{vmatrix} a & 0 & b \\ 0 & c & 0 \\ c & 0 & d \end{vmatrix} \\ & = -ad(ad - bc) + bc(ad - bc) \\ & = -a^2d^2 + 2abcd - b^2c^2 = -(ad - bc)^2, \end{aligned}$$

应选(B).

(6) **【答案】** (A).

【解】 若 $\alpha_1, \alpha_2, \alpha_3$ 线性无关,

$$\text{由 } (\alpha_1 + k\alpha_3, \alpha_2 + l\alpha_3) = (\alpha_1, \alpha_2, \alpha_3) \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ k & l \end{pmatrix},$$

因为 $(\alpha_1, \alpha_2, \alpha_3)$ 可逆, 所以 $\alpha_1 + k\alpha_3, \alpha_2 + l\alpha_3$ 的秩与矩阵 $\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ k & l \end{pmatrix}$ 的秩相等, 因为

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ k & l \end{pmatrix} \text{ 两列不成比例, 所以 } r \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ k & l \end{pmatrix} = 2, \text{ 故 } \alpha_1 + k\alpha_3, \alpha_2 + l\alpha_3 \text{ 线性无关.}$$

反之, 若 $\alpha_1 + k\alpha_3, \alpha_2 + l\alpha_3$ 线性无关, $\alpha_1, \alpha_2, \alpha_3$ 不一定线性无关,

如 α_1, α_2 线性无关, $\alpha_3 = \mathbf{0}$, 显然 $\alpha_1 + k\alpha_3, \alpha_2 + l\alpha_3$ 线性无关, 但 $\alpha_1, \alpha_2, \alpha_3$ 线性相关, 应选(A).

(7) **【答案】** (B).

【解】 由 $P(B) = 0.5$, 得 $P(\bar{B}) = 0.5$, 由 A, B 相互独立及减法公式得

$$P(A - B) = P(A\bar{B}) = P(A)P(\bar{B}) = 0.5P(A) = 0.3,$$

则 $P(A) = 0.6$, 从而 $P(\bar{A}) = 0.4$,

于是 $P(B - A) = P(\bar{A}B) = P(\bar{A})P(B) = 0.4 \times 0.5 = 0.2$, 应选(B).

(8) **【答案】** (C).

【解】 由 $X_1 - X_2 \sim N(0, 2\sigma^2)$ 得 $\frac{X_1 - X_2}{\sqrt{2}\sigma} \sim N(0, 1)$,

由 $X_3 \sim N(0, \sigma^2)$ 得 $\frac{X_3}{\sigma} \sim N(0, 1)$, 从而 $\frac{X_3^2}{\sigma^2} \sim \chi^2(1)$,

由 $\frac{X_1 - X_2}{\sqrt{2}\sigma}$ 与 $\frac{X_3^2}{\sigma^2}$ 独立得 $\frac{\frac{X_1 - X_2}{\sqrt{2}\sigma}}{\sqrt{\frac{X_3^2}{\sigma^2}/1}} \sim t(1)$, 即 $\frac{X_1 - X_2}{\sqrt{2} |X_3|} \sim t(1)$, 应选 (C).

二、填空题

(9) 【答案】 $20 - Q$.

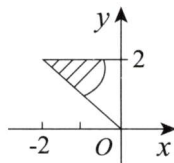
【解】 收益函数为 $R(Q) = pQ = Q\left(20 - \frac{Q}{2}\right) = 20Q - \frac{Q^2}{2}$,

该商品的边际收益为 $R'(Q) = 20 - Q$.

(10) 【答案】 $\frac{3}{2} - \ln 2$.

【解】 方法一 $A = \int_{-1}^{-\frac{1}{2}} \left(2 + \frac{1}{x}\right) dx + \int_{-2}^{-1} (2 + x) dx = \frac{3}{2} - \ln 2$.

方法二 $A = \int_1^2 dy \int_{-y}^{-\frac{1}{y}} dx = \frac{3}{2} - \ln 2$.



二(10)题图

(11) 【答案】 $\frac{1}{2}$.

【解】 由 $\int_0^a x e^{2x} dx = \frac{1}{4} \int_0^{2a} x e^x dx = \frac{1}{4} (x - 1) e^x \Big|_0^{2a} = \frac{1}{4} [(2a - 1) e^{2a} + 1] = \frac{1}{4}$,

得 $a = \frac{1}{2}$.

(12) 【答案】 $\frac{e-1}{2}$.

【解】
$$\begin{aligned} \int_0^1 dy \int_y^1 \left(\frac{e^{x^2}}{x} - e^{y^2} \right) dx &= \int_0^1 dy \int_y^1 \frac{e^{x^2}}{x} dx - \int_0^1 dy \int_y^1 e^{y^2} dx \\ &= \int_0^1 \frac{e^{x^2}}{x} dx \int_0^x dy - \int_0^1 (1-y) e^{y^2} dy = \int_0^1 e^{x^2} dx - \int_0^1 e^{y^2} dy + \int_0^1 y e^{y^2} dy \\ &= \int_0^1 y e^{y^2} dy = \frac{1}{2} e^{y^2} \Big|_0^1 = \frac{e-1}{2}. \end{aligned}$$

(13) 【答案】 $[-2, 2]$.

【解】 $A = \begin{pmatrix} 1 & 0 & a \\ 0 & -1 & 2 \\ a & 2 & 0 \end{pmatrix}$, $|A| = a^2 - 4$.

因为 A 的负惯性指数为 1, 所以 $|A| \leq 0$.

由 $|A| < 0$ 得 $-2 < a < 2$.

若 $|A| = 0$ 得 $a = -2$ 或 $a = 2$,

当 $a = -2$ 时, 由 $|\lambda E - A| = 0$ 得 $\lambda_1 = -3, \lambda_2 = 0, \lambda_3 = 3$, 负惯性指数为 1;

当 $a = 2$ 时, 由 $|\lambda E - A| = 0$ 得 $\lambda_1 = -3, \lambda_2 = 0, \lambda_3 = 3$, 负惯性指数为 1, 故 $-2 \leq a \leq 2$.

(14) 【答案】 $\frac{2}{5n}$.

【解】 $E(X^2) = \frac{2}{3\theta^2} \int_{\theta}^{2\theta} x^3 dx = \frac{5}{2}\theta^2$,

由 $E(c \sum_{i=1}^n X_i^2) = \frac{5nc}{2}\theta^2 = \theta^2$ 得 $c = \frac{2}{5n}$.

三、解答题

(15) 【解】 方法一

$$\begin{aligned} \lim_{x \rightarrow +\infty} \frac{\int_1^x [t^2(e^{\frac{1}{t}} - 1) - t] dt}{x^2 \ln\left(1 + \frac{1}{x}\right)} &= \lim_{x \rightarrow +\infty} \frac{\int_1^x [t^2(e^{\frac{1}{t}} - 1) - t] dt}{x} \cdot \frac{\frac{1}{x}}{\ln\left(1 + \frac{1}{x}\right)} \\ &= \lim_{x \rightarrow +\infty} \frac{\int_1^x [t^2(e^{\frac{1}{t}} - 1) - t] dt}{x} = \lim_{x \rightarrow +\infty} [x^2(e^{\frac{1}{x}} - 1) - x] \\ &= \lim_{x \rightarrow +\infty} x^2 \left(e^{\frac{1}{x}} - 1 - \frac{1}{x} \right) \stackrel{\frac{1}{x} = t}{=} \lim_{t \rightarrow 0} \frac{e^t - 1 - t}{t^2} \\ &= \lim_{t \rightarrow 0} \frac{e^t - 1}{2t} = \frac{1}{2}. \end{aligned}$$

方法二

$$\begin{aligned} \lim_{x \rightarrow +\infty} \frac{\int_1^x [t^2(e^{\frac{1}{t}} - 1) - t] dt}{x^2 \ln\left(1 + \frac{1}{x}\right)} &= \lim_{x \rightarrow +\infty} \frac{\int_1^x [t^2(e^{\frac{1}{t}} - 1) - t] dt}{x} \\ &= \lim_{x \rightarrow +\infty} [x^2(e^{\frac{1}{x}} - 1) - x] = \lim_{x \rightarrow +\infty} \left[x^2 \left(\frac{1}{x} + \frac{1}{2x^2} + o\left(\frac{1}{x^2}\right) \right) - x \right] \\ &= \lim_{x \rightarrow +\infty} x^2 \left[\frac{1}{2x^2} + o\left(\frac{1}{x^2}\right) \right] = \frac{1}{2}. \end{aligned}$$

(16) 【解】 方法一 由对称性得

$$I = \iint_D \frac{x \sin(\pi \sqrt{x^2 + y^2})}{x + y} dx dy = \iint_D \frac{y \sin(\pi \sqrt{x^2 + y^2})}{x + y} dx dy,$$

$$\begin{aligned} \text{于是 } I &= \frac{1}{2} \left[\iint_D \frac{x \sin(\pi \sqrt{x^2 + y^2})}{x + y} dx dy + \iint_D \frac{y \sin(\pi \sqrt{x^2 + y^2})}{x + y} dx dy \right] \\ &= \frac{1}{2} \iint_D \sin(\pi \sqrt{x^2 + y^2}) dx dy = \frac{1}{2} \int_0^{\frac{\pi}{2}} d\theta \int_1^2 r \sin \pi r dr = \frac{1}{4\pi} \int_1^2 \pi r \sin \pi r d(\pi r) \\ &= \frac{1}{4\pi} \int_{\pi}^{2\pi} t \sin t dt = -\frac{1}{4\pi} \int_{\pi}^{2\pi} t d(\cos t) = -\frac{1}{4\pi} t \cos t \Big|_{\pi}^{2\pi} + \frac{1}{4\pi} \int_{\pi}^{2\pi} \cos t dt = -\frac{3}{4}. \end{aligned}$$

方法二 $I = \iint_D \frac{x \sin(\pi \sqrt{x^2 + y^2})}{x + y} dx dy = \int_0^{\frac{\pi}{2}} \frac{\cos \theta}{\sin \theta + \cos \theta} d\theta \int_1^2 r \sin \pi r dr,$

因为 $\int_0^{\frac{\pi}{2}} \frac{\cos \theta}{\sin \theta + \cos \theta} d\theta = \int_0^{\frac{\pi}{2}} \frac{\sin \theta}{\sin \theta + \cos \theta} d\theta,$

$$\text{所以} \int_0^{\frac{\pi}{2}} \frac{\cos \theta}{\sin \theta + \cos \theta} d\theta = \frac{1}{2} \int_0^{\frac{\pi}{2}} \frac{\sin \theta + \cos \theta}{\sin \theta + \cos \theta} d\theta = \frac{\pi}{4},$$

$$\begin{aligned} \text{又因为} \int_1^2 r \sin \pi r dr &= \frac{1}{\pi^2} \int_{\pi}^{2\pi} t \sin t dt = -\frac{1}{\pi^2} \int_{\pi}^{2\pi} t d(\cos t) \\ &= -\frac{1}{\pi^2} t \cos t \Big|_{\pi}^{2\pi} + \frac{1}{\pi^2} \int_{\pi}^{2\pi} \cos t dt = -\frac{3}{\pi}, \end{aligned}$$

$$\text{所以} I = \iint_D \frac{x \sin(\pi \sqrt{x^2 + y^2})}{x + y} dx dy = -\frac{3}{4}.$$

$$\text{方法三} \quad I = \iint_D \frac{x \sin(\pi \sqrt{x^2 + y^2})}{x + y} dx dy = \int_0^{\frac{\pi}{2}} \frac{\cos \theta}{\sin \theta + \cos \theta} d\theta \int_1^2 r \sin \pi r dr,$$

$$\begin{aligned} \text{因为} \int_0^{\frac{\pi}{2}} \frac{\cos \theta}{\sin \theta + \cos \theta} d\theta &= \int_0^{\frac{\pi}{2}} \frac{1}{1 + \tan \theta} d\theta \\ &\stackrel{\tan \theta = t}{=} \int_0^{+\infty} \frac{1}{1+t} \cdot \frac{1}{1+t^2} dt = \frac{1}{2} \int_0^{+\infty} \left(\frac{1}{1+t} - \frac{t-1}{1+t^2} \right) dt \\ &= \frac{1}{2} \left[\ln(1+t) - \frac{1}{2} \ln(1+t^2) + \arctan t \right] \Big|_0^{+\infty} \\ &= \frac{1}{2} \ln \frac{1+t}{\sqrt{1+t^2}} \Big|_0^{+\infty} + \frac{1}{2} \arctan t \Big|_0^{+\infty} = \frac{\pi}{4}, \end{aligned}$$

$$\int_1^2 r \sin \pi r dr = \frac{1}{\pi^2} \int_{\pi}^{2\pi} t \sin t dt = -\frac{1}{\pi^2} \int_{\pi}^{2\pi} t d(\cos t) = -\frac{1}{\pi^2} t \cos t \Big|_{\pi}^{2\pi} + \frac{1}{\pi^2} \int_{\pi}^{2\pi} \cos t dt = -\frac{3}{\pi},$$

$$\text{所以} I = \iint_D \frac{x \sin(\pi \sqrt{x^2 + y^2})}{x + y} dx dy = -\frac{3}{4}.$$

$$\text{方法四} \quad I = \iint_D \frac{x \sin(\pi \sqrt{x^2 + y^2})}{x + y} dx dy = \int_0^{\frac{\pi}{2}} \frac{\cos \theta}{\sin \theta + \cos \theta} d\theta \int_1^2 r \sin \pi r dr,$$

$$\begin{aligned} \text{因为} \int_0^{\frac{\pi}{2}} \frac{\cos \theta}{\sin \theta + \cos \theta} d\theta &= \frac{1}{\sqrt{2}} \int_0^{\frac{\pi}{2}} \frac{\cos \left[\left(\theta + \frac{\pi}{4} \right) - \frac{\pi}{4} \right]}{\sin \left(\theta + \frac{\pi}{4} \right)} d\theta \\ &= \frac{1}{2} \int_0^{\frac{\pi}{2}} \frac{\cos \left(\theta + \frac{\pi}{4} \right) + \sin \left(\theta + \frac{\pi}{4} \right)}{\sin \left(\theta + \frac{\pi}{4} \right)} d\theta \\ &= \frac{1}{2} \int_0^{\frac{\pi}{2}} \left[1 + \cot \left(\theta + \frac{\pi}{4} \right) \right] d\theta \\ &= \frac{1}{2} \left[\theta + \ln \left| \sin \left(\theta + \frac{\pi}{4} \right) \right| \right] \Big|_0^{\frac{\pi}{2}} = \frac{\pi}{4}, \end{aligned}$$

$$\begin{aligned} \int_1^2 r \sin \pi r dr &= \frac{1}{\pi^2} \int_{\pi}^{2\pi} t \sin t dt = -\frac{1}{\pi^2} \int_{\pi}^{2\pi} t d(\cos t) \\ &= -\frac{1}{\pi^2} t \cos t \Big|_{\pi}^{2\pi} + \frac{1}{\pi^2} \int_{\pi}^{2\pi} \cos t dt = -\frac{3}{\pi}, \end{aligned}$$

$$\text{所以 } I = \iint_D \frac{x \sin(\pi \sqrt{x^2 + y^2})}{x + y} dx dy = -\frac{3}{4}.$$

$$\text{方法五 } I = \iint_D \frac{x \sin(\pi \sqrt{x^2 + y^2})}{x + y} dx dy = \int_0^{\frac{\pi}{2}} \frac{\cos \theta}{\sin \theta + \cos \theta} d\theta \int_1^2 r \sin \pi r dr,$$

$$\begin{aligned} \text{因为 } \int_0^{\frac{\pi}{2}} \frac{\cos \theta}{\sin \theta + \cos \theta} d\theta &= \frac{1}{2} \left[\int_0^{\frac{\pi}{2}} \frac{\cos \theta + \sin \theta}{\sin \theta + \cos \theta} d\theta + \int_0^{\frac{\pi}{2}} \frac{\cos \theta - \sin \theta}{\sin \theta + \cos \theta} d\theta \right] \\ &= \frac{1}{2} \left(\frac{\pi}{2} + \ln |\sin \theta + \cos \theta| \right) \Big|_0^{\frac{\pi}{2}} = \frac{\pi}{4}, \end{aligned}$$

$$\begin{aligned} \int_1^2 r \sin \pi r dr &= \frac{1}{\pi^2} \int_{\pi}^{2\pi} t \sin t dt = -\frac{1}{\pi^2} \int_{\pi}^{2\pi} t d(\cos t) \\ &= -\frac{1}{\pi^2} t \cos t \Big|_{\pi}^{2\pi} + \frac{1}{\pi^2} \int_{\pi}^{2\pi} \cos t dt = -\frac{3}{\pi}, \end{aligned}$$

$$\text{所以 } I = \iint_D \frac{x \sin(\pi \sqrt{x^2 + y^2})}{x + y} dx dy = -\frac{3}{4}.$$

$$(17) \text{ 【解】 } \frac{\partial z}{\partial x} = e^x \cos y \cdot f', \quad \frac{\partial z}{\partial y} = -e^x \sin y \cdot f',$$

$$\text{则 } \cos y \frac{\partial z}{\partial x} - \sin y \frac{\partial z}{\partial y} = (4z + e^x \cos y) e^x \text{ 化为 } f'(e^x \cos y) = 4z + e^x \cos y,$$

令 $u = e^x \cos y$, 则 $f'(u) - 4f(u) = u$, 解得

$$f(u) = \left(\int u e^{\int -4du} du + C \right) e^{-\int -4du} = C e^{4u} - \frac{u}{4} - \frac{1}{16},$$

$$\text{由 } f(0) = 0, \text{ 解得 } C = \frac{1}{16}, \text{ 故 } f(u) = \frac{1}{16} e^{4u} - \frac{1}{4} u - \frac{1}{16}.$$

方法点评:偏导数与微分方程结合问题是一种综合和重要的题型,首先按题目要求计算出相应的偏导数,根据给定的等量关系式将偏导数代入等式中,整理得微分方程,再根据微分方程的类型对微分方程求解.

$$(18) \text{ 【解】 } \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 1 \text{ 得幂级数的收敛半径为 } R = 1.$$

当 $x = \pm 1$ 时, 因为 $(n+1)(n+3) \rightarrow \infty \neq 0 (n \rightarrow \infty)$,

所以当 $x = \pm 1$ 时, 幂级数发散, 故幂级数的收敛域为 $(-1, 1)$.

$$\text{令 } S(x) = \sum_{n=0}^{\infty} (n+1)(n+3)x^n, \text{ 则}$$

$$\begin{aligned} S(x) &= \sum_{n=0}^{\infty} (n^2 + 4n + 3)x^n = \sum_{n=0}^{\infty} [(n+1)(n+2) + (n+1)]x^n \\ &= \sum_{n=0}^{\infty} (n+1)(n+2)x^n + \sum_{n=0}^{\infty} (n+1)x^n = \left(\sum_{n=0}^{\infty} x^{n+2} \right)' + \left(\sum_{n=0}^{\infty} x^{n+1} \right)' \\ &= \left(\frac{x^2}{1-x} \right)'' + \left(\frac{x}{1-x} \right)' = \frac{3-x}{(1-x)^3} (-1 < x < 1). \end{aligned}$$

$$(19) \text{ 【证明】 (I) 方法一 } \text{ 因为 } 0 \leq g(x) \leq 1,$$

$$\text{所以 } \int_a^x 0 dt \leq \int_a^x g(t) dt \leq \int_a^x 1 dt, \text{ 即 } 0 \leq \int_a^x g(t) dt \leq x - a, x \in [a, b].$$

方法二 令 $\varphi(x) = x - a - \int_a^x g(t) dt$,

因为 $\varphi'(x) = 1 - g(x) \geq 0$, 所以 $\varphi(x)$ 在 $[a, b]$ 上单调增加,

又因为 $\varphi(a) = 0$, 所以当 $x \in [a, b]$ 时, $\varphi(x) \geq 0$, 即 $\int_a^x g(t) dt \leq x - a$,

因为 $g(x) \geq 0$, 所以 $\int_a^x g(t) dt \geq 0$, 故当 $x \in [a, b]$ 时, $0 \leq \int_a^x g(t) dt \leq x - a$.

方法三 由积分中值定理得

$$\int_a^x g(t) dt = g(\xi)(x - a), \xi \in [a, x],$$

因为 $0 \leq g(x) \leq 1$, 所以当 $x \in [a, b]$ 时, $0 \leq \int_a^x g(t) dt \leq x - a$.

方法四 因为 $g(x)$ 在 $[0, 1]$ 上连续, 所以 $g(x)$ 在 $[0, 1]$ 上取到最小值 m 和最大值 M ,

因为 $0 \leq g(x) \leq 1$, 所以 $0 \leq m \leq g(x) \leq M \leq 1$, 积分得

$$0 \leq m(x - a) \leq \int_a^x g(t) dt \leq M(x - a) \leq x - a.$$

$$(\text{II}) \varphi(x) = \int_a^x f(u)g(u)du - \int_a^{a+\int_a^x g(t)dt} f(u)du,$$

因为 $f(x), g(x)$ 在 $[a, b]$ 上连续, 所以 $\varphi(x)$ 在 $[a, b]$ 上可导, 且

$$\varphi'(x) = f(x)g(x) - g(x)f\left[a + \int_a^x g(t)dt\right],$$

因为 $\int_a^x g(t)dt \leq x - a$ 且 $f(x)$ 单调增加,

$$\text{所以 } f\left[a + \int_a^x g(t)dt\right] \leq f[a + (x - a)] = f(x),$$

从而 $\varphi'(x) \geq f(x)g(x) - g(x)f(x) = 0, x \in [a, b]$,

由 $\begin{cases} \varphi(a) = 0, \\ \varphi'(x) \geq 0 (a \leq x \leq b) \end{cases}$ 得 $\varphi(x) \geq 0 (x \in [a, b])$,

从而 $\varphi(b) \geq 0$, 故 $\int_a^{a+\int_a^b g(t)dt} f(x)dx \leq \int_a^b f(x)g(x)dx$.

$$\begin{aligned} (20) \text{ 【解】 } (\text{I}) \mathbf{A} &= \begin{pmatrix} 1 & -2 & 3 & -4 \\ 0 & 1 & -1 & 1 \\ 1 & 2 & 0 & -3 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -2 & 3 & -4 \\ 0 & 1 & -1 & 1 \\ 0 & 4 & -3 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -2 & 3 & -4 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & -3 \end{pmatrix} \\ &\rightarrow \begin{pmatrix} 1 & -2 & 0 & 5 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & -3 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & -3 \end{pmatrix}, \end{aligned}$$

则方程组 $\mathbf{AX} = \mathbf{0}$ 的一个基础解系为 $\xi = (-1, 2, 3, 1)^T$.

(II) 方法一

$$\begin{aligned} \text{由 } (\mathbf{A} \vdots \mathbf{E}) &= \left(\begin{array}{cccc|ccc} 1 & -2 & 3 & -4 & 1 & 0 & 0 \\ 0 & 1 & -1 & 1 & 0 & 1 & 0 \\ 1 & 2 & 0 & -3 & 0 & 0 & 1 \end{array} \right) \rightarrow \left(\begin{array}{cccc|ccc} 1 & -2 & 3 & -4 & 1 & 0 & 0 \\ 0 & 1 & -1 & 1 & 0 & 1 & 0 \\ 0 & 4 & -3 & 1 & -1 & 0 & 1 \end{array} \right) \\ &\rightarrow \left(\begin{array}{cccc|ccc} 1 & -2 & 3 & -4 & 1 & 0 & 0 \\ 0 & 1 & -1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -3 & -1 & -4 & 1 \end{array} \right) \rightarrow \left(\begin{array}{cccc|ccc} 1 & -2 & 0 & 5 & 4 & 12 & -3 \\ 0 & 1 & 0 & -2 & -1 & -3 & 1 \\ 0 & 0 & 1 & -3 & -1 & -4 & 1 \end{array} \right) \end{aligned}$$

$$\rightarrow \left(\begin{array}{cccc|ccc} 1 & 0 & 0 & 1 & 2 & 6 & -1 \\ 0 & 1 & 0 & -2 & -1 & -3 & 1 \\ 0 & 0 & 1 & -3 & -1 & -4 & 1 \end{array} \right),$$

$$\text{得 } B = \begin{pmatrix} 2-k_1 & 6-k_2 & -1-k_3 \\ 2k_1-1 & 2k_2-3 & 2k_3+1 \\ 3k_1-1 & 3k_2-4 & 3k_3+1 \\ k_1 & k_2 & k_3 \end{pmatrix} \quad (k_1, k_2, k_3 \text{ 为任意常数}).$$

方法二 令 $B = (X_1, X_2, X_3)$, $E = (e_1, e_2, e_3)$,

则 $AB = E$ 等价于 $AX_1 = e_1$, $AX_2 = e_2$, $AX_3 = e_3$,

$$\text{方程组 } AX_1 = e_1 \text{ 的通解为 } X_1 = k_1 \begin{pmatrix} -1 \\ 2 \\ 3 \\ 1 \end{pmatrix} + \begin{pmatrix} 2 \\ -1 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} -k_1+2 \\ 2k_1-1 \\ 3k_1-1 \\ k_1 \end{pmatrix} \quad (k_1 \text{ 为任意常数}),$$

$$\text{方程组 } AX_2 = e_2 \text{ 的通解为 } X_2 = k_2 \begin{pmatrix} -1 \\ 2 \\ 3 \\ 1 \end{pmatrix} + \begin{pmatrix} 6 \\ -3 \\ -4 \\ 0 \end{pmatrix} = \begin{pmatrix} -k_2+6 \\ 2k_2-3 \\ 3k_2-4 \\ k_2 \end{pmatrix} \quad (k_2 \text{ 为任意常数}),$$

$$\text{方程组 } AX_3 = e_3 \text{ 的通解为 } X_3 = k_3 \begin{pmatrix} -1 \\ 2 \\ 3 \\ 1 \end{pmatrix} + \begin{pmatrix} -1 \\ 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -k_3-1 \\ 2k_3+1 \\ 3k_3+1 \\ k_3 \end{pmatrix} \quad (k_3 \text{ 为任意常数}),$$

$$\text{故 } B = \begin{pmatrix} -k_1+2 & -k_2+6 & -k_3-1 \\ 2k_1-1 & 2k_2-3 & 2k_3+1 \\ 3k_1-1 & 3k_2-4 & 3k_3+1 \\ k_1 & k_2 & k_3 \end{pmatrix} \quad (k_1, k_2, k_3 \text{ 为任意常数}).$$

$$\text{方法三 令 } B = \begin{pmatrix} x_1 & x_2 & x_3 \\ x_4 & x_5 & x_6 \\ x_7 & x_8 & x_9 \\ x_{10} & x_{11} & x_{12} \end{pmatrix}, \text{ 由 } AB = E \text{ 得}$$

$$\begin{cases} x_1 - 2x_4 + 3x_7 - 4x_{10} = 1, \\ x_2 - 2x_5 + 3x_8 - 4x_{11} = 0, \\ x_3 - 2x_6 + 3x_9 - 4x_{12} = 0, \\ x_4 - x_7 + x_{10} = 0, \\ x_5 - x_8 + x_{11} = 1, \\ x_6 - x_9 + x_{12} = 0, \\ x_1 + 2x_4 - 3x_{10} = 0, \\ x_2 + 2x_5 - 3x_{11} = 0, \\ x_3 + 2x_6 - 3x_{12} = 1, \end{cases}$$

$$\text{解得} \begin{pmatrix} x_1 \\ x_4 \\ x_7 \\ x_{10} \end{pmatrix} = k_1 \begin{pmatrix} -1 \\ 2 \\ 3 \\ 1 \end{pmatrix} + \begin{pmatrix} 2 \\ -1 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} -k_1 + 2 \\ 2k_1 - 1 \\ 3k_1 - 1 \\ k_1 \end{pmatrix},$$

$$\begin{pmatrix} x_2 \\ x_5 \\ x_8 \\ x_{11} \end{pmatrix} = k_2 \begin{pmatrix} -1 \\ 2 \\ 3 \\ 1 \end{pmatrix} + \begin{pmatrix} 6 \\ -3 \\ -4 \\ 0 \end{pmatrix} = \begin{pmatrix} -k_2 + 6 \\ 2k_2 - 3 \\ 3k_2 - 4 \\ k_2 \end{pmatrix},$$

$$\begin{pmatrix} x_3 \\ x_6 \\ x_9 \\ x_{12} \end{pmatrix} = k_3 \begin{pmatrix} -1 \\ 2 \\ 3 \\ 1 \end{pmatrix} + \begin{pmatrix} -1 \\ 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -k_3 - 1 \\ 2k_3 + 1 \\ 3k_3 + 1 \\ k_3 \end{pmatrix},$$

$$\text{故 } \mathbf{B} = \begin{pmatrix} -k_1 + 2 & -k_2 + 6 & -k_3 - 1 \\ 2k_1 - 1 & 2k_2 - 3 & 2k_3 + 1 \\ 3k_1 - 1 & 3k_2 - 4 & 3k_3 + 1 \\ k_1 & k_2 & k_3 \end{pmatrix} \quad (k_1, k_2, k_3 \text{ 为任意常数}).$$

方法点评:求未知矩阵一般有如下三种情形:

(1) 矩阵方程经过化简得 $\mathbf{AX} = \mathbf{B}$ 或 $\mathbf{XA} = \mathbf{B}$, 其中 $\mathbf{A}$ 可逆, 则 $\mathbf{X} = \mathbf{A}^{-1}\mathbf{B}$ 或 $\mathbf{X} = \mathbf{BA}^{-1}$;

(2) 矩阵方程经过化简得 $\mathbf{AX} = \mathbf{B}$, 其中 $\mathbf{A}$ 不可逆或 $\mathbf{A}$ 不是方阵, 则一般将 $\mathbf{AX} = \mathbf{B}$ 拆成几个方程组, 求每个方程组的通解, 将通解合成矩阵 $\mathbf{X}$;

(3) 矩阵对角化法

设 $\mathbf{A}$ 的特征值为 $\lambda_1, \lambda_2, \dots, \lambda_n$, 其对应的线性无关的特征向量为 $\alpha_1, \alpha_2, \dots, \alpha_n$,

$$\text{令 } \mathbf{P} = (\alpha_1, \alpha_2, \dots, \alpha_n), \mathbf{P} \text{ 可逆, 且 } \mathbf{P}^{-1}\mathbf{AP} = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix},$$

$$\text{于是 } \mathbf{A} = \mathbf{P} \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} \mathbf{P}^{-1}.$$

$$(21) \text{ 【证明】 令 } \mathbf{A} = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & \vdots & & \vdots \\ 1 & 1 & \cdots & 1 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 0 & \cdots & 0 & 1 \\ 0 & \cdots & 0 & 2 \\ \vdots & & \vdots & \vdots \\ 0 & \cdots & 0 & n \end{pmatrix},$$

由 $|\lambda \mathbf{E} - \mathbf{A}| = 0$ 得 $\mathbf{A}$ 的特征值为 $\lambda_1 = \cdots = \lambda_{n-1} = 0, \lambda_n = n$,

由 $|\lambda \mathbf{E} - \mathbf{B}| = 0$ 得 $\mathbf{B}$ 的特征值为 $\lambda_1 = \cdots = \lambda_{n-1} = 0, \lambda_n = n$.

因为 $\mathbf{A}^T = \mathbf{A}$, 所以 $\mathbf{A}$ 可对角化;

因为 $r(0\mathbf{E} - \mathbf{B}) = r(\mathbf{B}) = 1$, 所以 $\mathbf{B}$ 可对角化,

因为 $\mathbf{A}, \mathbf{B}$ 特征值相同且都可对角化, 所以 $\mathbf{A} \sim \mathbf{B}$.

方法点评: 本题考查矩阵相似.

设 A, B 为两个 n 阶矩阵, 若 $A \sim B$, 则 A, B 的特征值相同; 反之, 若 A, B 的特征值相同, 两矩阵不一定相似, 即特征值相同是两个矩阵相似的必要而非充分条件.

注意如下结论:

(1) 若 A, B 特征值相同, 且 A, B 都可相似对角化, 则 $A \sim B$;

(2) 若 A, B 特征值相同, 但 A, B 中一个可相似对角化, 另一个不可相似对角化, 则两矩阵一定不相似.

(22) 【解】 (I)

$$\begin{aligned} F_Y(y) &= P\{Y \leq y\} = P\{X=1\}P\{Y \leq y \mid X=1\} + P\{X=2\}P\{Y \leq y \mid X=2\} \\ &= \frac{1}{2}P\{Y \leq y \mid X=1\} + \frac{1}{2}P\{Y \leq y \mid X=2\}, \end{aligned}$$

当 $y < 0$ 时, $F_Y(y) = 0$;

$$\text{当 } 0 \leq y < 1 \text{ 时, } F_Y(y) = \frac{1}{2} \cdot \frac{y}{1} + \frac{1}{2} \cdot \frac{y}{2} = \frac{3y}{4};$$

$$\text{当 } 1 \leq y < 2 \text{ 时, } F_Y(y) = \frac{1}{2} + \frac{1}{2} \cdot \frac{y}{2} = \frac{y}{4} + \frac{1}{2};$$

当 $y \geq 2$ 时, $F_Y(y) = 1$,

$$\text{故随机变量 } Y \text{ 的分布函数为 } F_Y(y) = \begin{cases} 0, & y < 0, \\ \frac{3y}{4}, & 0 \leq y < 1, \\ \frac{y}{4} + \frac{1}{2}, & 1 \leq y < 2, \\ 1, & y \geq 2. \end{cases}$$

$$\text{(II) 随机变量 } Y \text{ 的密度函数为 } f_Y(y) = \begin{cases} \frac{3}{4}, & 0 < y < 1, \\ \frac{1}{4}, & 1 < y < 2, \\ 0, & \text{其他.} \end{cases}$$

$$\text{则 } E(Y) = \int_0^1 \frac{3x}{4} dx + \int_1^2 \frac{x}{4} dx = \frac{3}{8} + \frac{3}{8} = \frac{3}{4}.$$

(23) 【解】 (I) 由 $X \sim \begin{pmatrix} 0 & 1 \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix}$ 得

$$E(X) = \frac{2}{3}, \quad E(X^2) = \frac{2}{3}, \quad D(X) = E(X^2) - [E(X)]^2 = \frac{2}{9},$$

$$\text{同理由 } Y \sim \begin{pmatrix} 0 & 1 \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix}, \text{ 得 } E(Y) = \frac{2}{3}, \quad D(Y) = \frac{2}{9},$$

$$\text{由 } \rho_{XY} = \frac{E(XY) - E(X)E(Y)}{\sqrt{D(X)} \cdot \sqrt{D(Y)}} = \frac{E(XY) - \frac{4}{9}}{\frac{2}{9}} = \frac{1}{2} \text{ 得 } E(XY) = \frac{5}{9},$$

$$\text{故 } P\{X=1, Y=1\} = \frac{5}{9},$$

$$\text{由 } P\{X=1, Y=0\} + P\{X=1, Y=1\} = \frac{2}{3} \text{ 得 } P\{X=1, Y=0\} = \frac{1}{9},$$

$$\text{由 } P\{X=0, Y=0\} + P\{X=1, Y=0\} = \frac{1}{3} \text{ 得 } P\{X=0, Y=0\} = \frac{2}{9},$$

$$\text{由 } P\{X=0, Y=1\} + P\{X=1, Y=1\} = \frac{2}{3} \text{ 得 } P\{X=0, Y=1\} = \frac{1}{9},$$

故 (X, Y) 的联合分布律为

X	Y		$P_{i\cdot}$
	0	1	
0	$\frac{2}{9}$	$\frac{1}{9}$	$\frac{1}{3}$
1	$\frac{1}{9}$	$\frac{5}{9}$	$\frac{2}{3}$
$P_{\cdot j}$	$\frac{1}{3}$	$\frac{2}{3}$	1

$$(\text{II}) P\{X+Y \leqslant 1\} = 1 - P\{X+Y > 1\} = 1 - P\{X=1, Y=1\} = 1 - \frac{5}{9} = \frac{4}{9}.$$